2.30 Integration: Advanced Methods (Hyperbolics)

Worked Solutions

1 a)

Using $\operatorname{cosech} x = 1/\sinh x$ and $\sinh x = (e^x - e^{-x})/2$,

$$\operatorname{cosech} x = \frac{2}{e^x - e^{-x}} = \frac{2e^x}{e^{2x} - 1}.$$

1 b)

Let $u = e^x$, so $dx = du/u$. Then

$$\int \operatorname{cosech} x \, dx = \int \frac{2}{u^2 - 1} \, du = -2 \int \frac{1}{1 - u^2} \, du = -2 \operatorname{artanh} u + c = -2 \operatorname{artanh}(e^x) + c.$$

2 a)

$$\operatorname{sech} x = \frac{1}{\cosh x} = \frac{2}{e^x + e^{-x}} = \frac{2e^x}{e^{2x} + 1}.$$

2 b)

Let $u = e^x$, so $dx = du/u$. Thus

$$\int \operatorname{sech} x \, dx = \int \frac{2}{u^2 + 1} \, du = 2 \arctan u + c = 2 \arctan(e^x) + c.$$

3)

First find where the curves intersect. Put $C = \cosh x$:

$$12 \operatorname{sech} x = 24 - 9 \cosh x \quad \Longrightarrow \quad \frac{4}{C} = 8 - 3C \quad \Longrightarrow \quad 3C^2 - 8C + 4 = 0 \quad \Longrightarrow \quad (3C - 2)(C - 2) = 0.$$

Since $\cosh x \geq 1$ for all real x, $C \neq 2/3$, so $C = 2$. Hence

$$x = \pm \ln(2 + \sqrt{2^2 - 1}) = \pm \ln(2 + \sqrt{3}), \qquad -\ln(2 + \sqrt{3}) = \ln(2 - \sqrt{3}).$$

Now integrate between these limits:

$$\begin{aligned} I &= \int_{\ln(2-\sqrt{3})}^{\ln(2+\sqrt{3})} (24 - 9 \cosh x) \, dx - \int_{\ln(2-\sqrt{3})}^{\ln(2+\sqrt{3})} 12 \operatorname{sech} x \, dx \\ &= \int_{\ln(2-\sqrt{3})}^{\ln(2+\sqrt{3})} (24 - 9 \cosh x - 12 \operatorname{sech} x) \, dx. \end{aligned}$$

As in Question 2, $\int \operatorname{sech} x \, dx = 2 \arctan(e^x) + c$, and therefore

$$I = \left[24x - 9 \sinh x - 24 \arctan(e^x) \right]_{\ln(2-\sqrt{3})}^{\ln(2+\sqrt{3})}.$$

Evaluating the limits as in the printed working,

$$I = \left(24 \ln(2 + \sqrt{3}) - \frac{9}{2} \left(e^{\ln(2+\sqrt{3})} - e^{-\ln(2+\sqrt{3})} \right) - 24 \cdot \frac{5\pi}{12} \right)$$

$$
\begin{aligned}
&\quad - \left(24\ln(2-\sqrt{3}) - \frac{9}{2}\left(e^{\ln(2-\sqrt{3})} - e^{-\ln(2-\sqrt{3})}\right) - 24\cdot\frac{\pi}{12}\right) \\
&= \left(24\ln(2+\sqrt{3}) - \frac{9}{2}\left((2+\sqrt{3}) - (2-\sqrt{3})\right) - 20\pi\right) \\
&\quad - \left(24\ln(2-\sqrt{3}) - \frac{9}{2}\left((2-\sqrt{3}) - (2+\sqrt{3})\right) - 2\pi\right) \\
&= \left(24\ln(2+\sqrt{3}) - 9\sqrt{3} - 10\pi\right) - \left(24\ln(2-\sqrt{3}) + 9\sqrt{3} - 2\pi\right) \\
&= 24\ln\left(\frac{2+\sqrt{3}}{2-\sqrt{3}}\right) - 18\sqrt{3} - 8\pi \\
&= 24\ln(7+4\sqrt{3}) - 18\sqrt{3} - 8\pi.
\end{aligned}
$$

4 a)

$$\int \tanh x\,dx = \int \frac{\sinh x}{\cosh x}\,dx = \ln|\cosh x| + c.$$

4 b)

$$\int \coth x\,dx = \int \frac{\cosh x}{\sinh x}\,dx = \ln|\sinh x| + c.$$

5 a)

$$
\begin{aligned}
\int \operatorname{cosech} x \coth x\,dx = \int \cosh x(\sinh x)^{-2}\,dx &= -(\sinh x)^{-1} + c \\
&= -\operatorname{cosech} x + c.
\end{aligned}
$$

Alternatively, writing in exponentials and using $u = e^x - e^{-x}$ gives

$$\int \frac{2e^x + 2e^{-x}}{(e^x - e^{-x})^2}\,dx = \int \frac{2}{u^2}\,du = -\frac{2}{u} + c = -\operatorname{cosech} x + c.$$

5 b)

$$
\begin{aligned}
\int \operatorname{sech} x \tanh x\,dx = \int \sinh x(\cosh x)^{-2}\,dx &= -(\cosh x)^{-1} + c \\
&= -\operatorname{sech} x + c.
\end{aligned}
$$

Alternatively, using $u = e^x + e^{-x}$ in the exponential form gives $\int 2u^{-2}\,du = -2/u + c = -\operatorname{sech} x + c$.

6 a)

$$\cosh^2 x - \sinh^2 x = \left(\frac{e^x + e^{-x}}{2}\right)^2 - \left(\frac{e^x - e^{-x}}{2}\right)^2 = \frac{e^{2x} + 2 + e^{-2x} - e^{2x} + 2 - e^{-2x}}{4} = 1.$$

6 b)

Using $\cosh^2 x - \sinh^2 x = 1$,

$$\begin{aligned}\int \operatorname{cosech}^2 x \, dx &= \int \frac{\cosh^2 x - \sinh^2 x}{\sinh^2 x} \, dx \\ &= -\int \frac{\sinh x \, d(\cosh x) - \cosh x \, d(\sinh x)}{\sinh^2 x} = -\coth x + c.\end{aligned}$$

Alternatively, writing $u = e^x$ and substituting $v = u^2 - 1$ gives

$$\int \left(\frac{2}{e^x - e^{-x}}\right)^2 dx = \int \frac{4u}{(u^2-1)^2} \, du = -\frac{2}{u^2-1} + c = -\coth x + c.$$

7 a)

$$y = \int \operatorname{cosech} 3x \coth 3x \, dx = -\frac{1}{3} \operatorname{cosech} 3x + c.$$

7 b)

$$y = 3\int \operatorname{cosech}^2(x/9) \, dx = -27 \coth(x/9) + c.$$

7 c)

$$y = 6\int \operatorname{sech} 9x \tanh 9x \, dx = -\frac{2}{3} \operatorname{sech} 9x + c.$$

7 d)

$$y = 10\int \coth 2x \, dx = 5 \ln|\sinh 2x| + c.$$

8)

Let $u = \sinh x$, so $du = \cosh x \, dx$. Then

$$\int \frac{\cosh x}{\sinh^2 x + 1} \, dx = \int \frac{1}{u^2+1} \, du = \arctan(\sinh x) + c.$$

9)

Let $u = \sin x$, so $du = \cos x \, dx$. Thus

$$\int \frac{\cos x}{\sqrt{\sin^2 x + 1}} \, dx = \int \frac{du}{\sqrt{u^2+1}} = \operatorname{arsinh} u + c = \operatorname{arsinh}(\sin x) + c.$$

10 a)

$$\frac{du}{dx} = 2\cosh x \sinh x = \sinh 2x.$$

10 b)

With $u = \cosh^2 x$, $du = \sinh 2x\, dx$, so

$$\int \frac{\sinh 2x}{\cosh^4 x + 1}\, dx = \int \frac{du}{u^2 + 1} = \arctan u + c = \arctan(\cosh^2 x) + c.$$

11 a)

Integration by parts with $u = \operatorname{arsinh} x$, $dv = dx$ gives

$$\int \operatorname{arsinh} x\, dx = x \operatorname{arsinh} x - \int \frac{x}{\sqrt{x^2+1}}\, dx = x \operatorname{arsinh} x - \sqrt{x^2+1} + c.$$

11 b)

Integration by parts gives

$$\int \operatorname{arcosh} x\, dx = x \operatorname{arcosh} x - \int \frac{x}{\sqrt{x^2-1}}\, dx = x \operatorname{arcosh} x - \sqrt{x^2-1} + c.$$

11 c)

Integration by parts gives $\int \operatorname{artanh} x\, dx = x \operatorname{artanh} x - \int \frac{x}{1-x^2}\, dx$. With $u = 1 - x^2$,

$$\int \frac{x}{1-x^2}\, dx = -\frac{1}{2} \ln|1 - x^2|, \qquad \int \operatorname{artanh} x\, dx = x \operatorname{artanh} x + \frac{1}{2} \ln|1 - x^2| + c.$$

12)

Integration by parts with $u = \operatorname{artanh} x$, $dv = 6\, dx$ gives

$$\int 6 \operatorname{artanh} x\, dx = 6x \operatorname{artanh} x + 3 \ln|1 - x^2| + c.$$

Therefore

$$\begin{aligned}\int_0^{1/3} 6 \operatorname{artanh} x\, dx &= \left[3x \ln\left(\frac{1+x}{1-x}\right) + 3 \ln|1 - x^2|\right]_0^{1/3} \\ &= \ln 2 + 3 \ln\left(\frac{8}{9}\right) = \ln 2 + \ln\left(\frac{2^9}{3^6}\right) = \ln\left(\frac{2^{10}}{3^6}\right).\end{aligned}$$

Hence $p = 10$ and $q = 6$.

13)

Integration by parts, with $u = \operatorname{artanh}(2x)$ and $dv = 4\, dx$, gives

$$\int 4 \operatorname{artanh}(2x)\, dx = 4x \operatorname{artanh}(2x) + \ln|1 - 4x^2| + c.$$

Thus

$$\begin{aligned}\int_0^{\tanh 4/2} 4 \operatorname{artanh}(2x)\, dx &= \left[2x \ln\left(\frac{1+2x}{1-2x}\right) + \ln|1 - 4x^2|\right]_0^{\tanh 4/2} \\ &= \tanh 4 \ln\left(\frac{1+\tanh 4}{1-\tanh 4}\right) + \ln|1 - \tanh^2 4| \\ &= 8 \tanh 4 + \ln|1 - \tanh^2 4|.\end{aligned}$$

14 a)

$$\frac{d}{dx}(x\,\mathrm{arcosh}\,3x) = \mathrm{arcosh}\,3x + \frac{3x}{\sqrt{9x^2-1}}.$$

14 b)

Using part a) and integration by parts,

$$\begin{aligned}\int \mathrm{arcosh}\,3x\,dx &= x\,\mathrm{arcosh}\,3x - \int \frac{3x}{\sqrt{9x^2-1}}\,dx \\ &= x\,\mathrm{arcosh}\,3x - \frac{1}{3}\sqrt{9x^2-1} + c,\end{aligned}$$

where the remaining integral follows from $u = 9x^2 - 1$.

15)

Integration by parts, with $u = \mathrm{arcosh}\,x$ and $dv = x^2\,dx$, gives

$$\int x^2\,\mathrm{arcosh}\,x\,dx = \frac{1}{3}x^3\,\mathrm{arcosh}\,x - \frac{1}{3}\int \frac{x^3}{\sqrt{x^2-1}}\,dx.$$

Put $u = x^2 - 1$, so $x^2 = u + 1$ and $du = 2x\,dx$. Then

$$\int \frac{x^3}{\sqrt{x^2-1}}\,dx = \frac{1}{2}\int (u+1)u^{-1/2}\,du = \frac{1}{3}u^{3/2} + u^{1/2} + c.$$

Hence

$$\int x^2\,\mathrm{arcosh}\,x\,dx = \frac{1}{3}x^3\,\mathrm{arcosh}\,x - \frac{1}{9}(x^2-1)^{3/2} - \frac{1}{3}\sqrt{x^2-1} + c.$$

16 a)

$$\frac{2x^2}{1-x^2} = \frac{2x^2-2+2}{1-x^2} = -2 + \frac{2}{1-x^2}, \qquad A = -2, \quad B = 2.$$

16 b)

By parts with $u = \mathrm{artanh}\,x$ and $dv = 4x\,dx$,

$$\begin{aligned}\int 4x\,\mathrm{artanh}\,x\,dx &= 2x^2\,\mathrm{artanh}\,x - \int \frac{2x^2}{1-x^2}\,dx \\ &= 2x^2\,\mathrm{artanh}\,x - \int \left(-2 + \frac{2}{1-x^2}\right) dx \\ &= 2x^2\,\mathrm{artanh}\,x + 2x - 2\,\mathrm{artanh}\,x + c.\end{aligned}$$

17 a)

Taking $u = \mathrm{arsinh}\,x$, $dv = dx/\sqrt{x^2+1}$, so $v = \mathrm{arsinh}\,x$, gives

$$I = (\mathrm{arsinh}\,x)^2 - I, \qquad I = \frac{1}{2}(\mathrm{arsinh}\,x)^2 + c.$$

17 b)

Similarly, with $u = \operatorname{arcosh} x$ and $dv = dx/\sqrt{x^2-1}$, $v = \operatorname{arcosh} x$:

$$I = (\operatorname{arcosh} x)^2 - I, \qquad I = \frac{1}{2}(\operatorname{arcosh} x)^2 + c.$$

17 c)

With $u = \operatorname{artanh} x$ and $dv = dx/(1-x^2)$, $v = \operatorname{artanh} x$:

$$I = (\operatorname{artanh} x)^2 - I, \qquad I = \frac{1}{2}(\operatorname{artanh} x)^2 + c.$$

18)

Integration by parts with $u = \operatorname{arsinh} 2x$ and $dv = dx/\sqrt{4x^2+1}$, so $v = \frac{1}{2}\operatorname{arsinh} 2x$, yields

$$\int \frac{\operatorname{arsinh} 2x}{\sqrt{4x^2+1}}\,dx = \frac{1}{4}(\operatorname{arsinh} 2x)^2 + c.$$

Therefore

$$\int_0^2 \frac{\operatorname{arsinh} 2x}{\sqrt{4x^2+1}}\,dx = \frac{1}{4}(\operatorname{arsinh} 4)^2 = \frac{1}{4}\left(\ln(4+\sqrt{17})\right)^2.$$

19)

Integration by parts gives

$$\int \frac{\operatorname{arcosh} 3x}{\sqrt{9x^2-1}}\,dx = \frac{1}{6}(\operatorname{arcosh} 3x)^2 + c.$$

Consequently,

$$\int_{1/3}^4 \frac{\operatorname{arcosh} 3x}{\sqrt{9x^2-1}}\,dx = \frac{1}{6}(\operatorname{arcosh} 12)^2 = \frac{1}{6}\left(\ln(12+\sqrt{143})\right)^2.$$

20 a)

$$\frac{10}{5\cosh x - 4\sinh x} = \frac{10}{\frac{5}{2}(e^x+e^{-x}) - 2(e^x - e^{-x})} = \frac{20}{e^x + 9e^{-x}} = \frac{20e^x}{e^{2x}+9}.$$

Thus $p = 20$ and $q = 9$.

20 b)

Let $u = e^x$, so the limits $x = 0$ and $x = \ln 3$ become $u = 1$ and $u = 3$:

$$\int_0^{\ln 3} \frac{30}{5\cosh x - 4\sinh x}\,dx = \int_1^3 \frac{60}{u^2+9}\,du = \left[20\arctan\left(\frac{u}{3}\right)\right]_1^3 = 5\pi - 20\arctan\frac{1}{3}.$$

21)

Put $u = \operatorname{artanh} x$, so $du = dx/(1-x^2)$. Hence

$$\int \frac{1}{(1-x^2)((\operatorname{artanh} x)^2+1)}\,dx = \int \frac{1}{u^2+1}\,du = \arctan(\operatorname{artanh} x) + c.$$

22 a)

$$\frac{dy}{dx} = \frac{1}{8\sqrt{x}} \frac{1}{\sqrt{x/16 - 1}} = \frac{1}{2\sqrt{x(x-16)}}.$$

22 b)

From part a), $2\,dy/dx = 1/\sqrt{x(x-16)}$. Therefore

$$\begin{aligned}\int_{64}^{144} \frac{1}{\sqrt{x(x-16)}}\,dx &= 2\left[\operatorname{arcosh}\left(\frac{\sqrt{x}}{4}\right)\right]_{64}^{144} \\ &= 2(\operatorname{arcosh} 3 - \operatorname{arcosh} 2) \\ &= 2\ln\left(\frac{3+2\sqrt{2}}{2+\sqrt{3}}\right).\end{aligned}$$

23)

Integration by parts with $u = \operatorname{arsinh} x$ and $dv = x\,dx/\sqrt{x^2+1}$, so $v = \sqrt{x^2+1}$, gives

$$\int \frac{x \operatorname{arsinh} x}{\sqrt{x^2+1}}\,dx = \sqrt{x^2+1}\,\operatorname{arsinh} x - \int 1\,dx = \sqrt{x^2+1}\,\operatorname{arsinh} x - x + c.$$

24 a)

$$\frac{d}{dx}(\operatorname{arcosh} 2x) = \frac{2}{\sqrt{4x^2-1}}.$$

24 b)

Set $u = \operatorname{arcosh} 2x$ and $dv = x\,dx/\sqrt{4x^2-1}$, so $v = \frac{1}{4}\sqrt{4x^2-1}$:

$$\int \frac{x \operatorname{arcosh} 2x}{\sqrt{4x^2-1}}\,dx = \frac{1}{4}\sqrt{4x^2-1}\,\operatorname{arcosh} 2x - \frac{1}{4}\int 2\,dx = \frac{1}{4}\sqrt{4x^2-1}\,\operatorname{arcosh} 2x - \frac{x}{2} + c.$$

25 a)

Using $\sinh(2u) = 2\sinh u \cosh u$, with $u = \operatorname{arsinh} x$, gives

$$\sinh(2\operatorname{arsinh} x) = 2x\cosh(\operatorname{arsinh} x) = 2x\sqrt{x^2+1}.$$

25 b)

Integration by parts with $u = \operatorname{arsinh} x$ and $dv = x\,dx$ gives

$$\begin{aligned}\int x \operatorname{arsinh} x\,dx &= \frac{x^2}{2}\operatorname{arsinh} x - \frac{1}{2}\int \frac{x^2}{\sqrt{x^2+1}}\,dx \\ &= \frac{x^2}{2}\operatorname{arsinh} x - \frac{1}{2}\int \left(\sqrt{x^2+1} - \frac{1}{\sqrt{x^2+1}}\right)dx \\ &= \frac{x^2-1}{2}\operatorname{arsinh} x + \frac{x\sqrt{x^2+1}}{2} + c.\end{aligned}$$

Here $\int \sqrt{x^2+1}\,dx = \frac{1}{2}(x\sqrt{x^2+1} + \operatorname{arsinh} x)$ and $\int dx/\sqrt{x^2+1} = \operatorname{arsinh} x$.

26 a)

Writing $\operatorname{cosech} x = 2/(e^x - e^{-x})$ and setting $u = e^x$ gives

$$\int \operatorname{cosech} x \, dx = \int \frac{2}{u^2 - 1} \, du = \ln \left| \frac{u-1}{u+1} \right| + c = \ln \left| \tanh \frac{x}{2} \right| + c.$$

26 b i)

As in Question 2, $u = e^x$ gives

$$\int \operatorname{sech} x \, dx = \int \frac{2}{u^2 + 1} \, du = 2 \arctan(e^x) + c.$$

26 b ii)

Using exponentials,

$$1 + \tanh \frac{x}{2} = \frac{2e^{x/2}}{e^{x/2} + e^{-x/2}}, \qquad 1 - \tanh \frac{x}{2} = \frac{2e^{-x/2}}{e^{x/2} + e^{-x/2}},$$

so their ratio is e^x. Alternatively, $\tanh(x/2) = (e^x - 1)/(e^x + 1)$; rearranging gives the same result.

26 b iii)

Let $t = \tanh(x/2)$. From part ii), $e^x = (1 + t)/(1 - t)$. The tangent addition formula gives

$$\tan \left(\frac{\pi}{4} + \arctan t \right) = \frac{1 + t}{1 - t} = e^x,$$

so $2 \arctan(e^x) = 2 \arctan(t) + \pi/2$. The constant $\pi/2$ is absorbed into c, and therefore

$$\int \operatorname{sech} x \, dx = 2 \arctan \left(\tanh \frac{x}{2} \right) + c.$$

27 a)

Writing $\operatorname{sech} x = 2/(e^x + e^{-x})$ and differentiating,

$$\frac{d}{dx}(\operatorname{sech} x) = -\frac{2(e^x - e^{-x})}{(e^x + e^{-x})^2} = -\operatorname{sech} x \tanh x.$$

27 b)

By parts with $u = x$ and $dv = \operatorname{sech} x \tanh x \, dx$, so $v = -\operatorname{sech} x$,

$$\begin{aligned} \int x \operatorname{sech} x \tanh x \, dx &= -x \operatorname{sech} x + \int \operatorname{sech} x \, dx \\ &= -x \operatorname{sech} x + 2 \arctan(e^x) + c. \end{aligned}$$